

Multiple Choice (10 marks)

1. If m and n are odd integers, how many terms in the expansion of $(m + n)^6$ are odd?
 - (a) 6
 - (b) 3
 - (c) All the terms are odd
 - (d) 4
 - (e) 1
2. If f is a linear function with nonzero slope, and $c < d$, which of the following must be FALSE?
 - (a) $f(d) > f(c)$
 - (b) $f(c) < f(d)$
 - (c) $f(d) \neq f(c)$
 - (d) $f(c) = f(d)$
 - (e) $f(d) < f(c)$
3. Statement 1 | Every solvable group is of prime-power order. Statement 2 | Every group of prime-power order is solvable.
 - (a) Statement 1 is always true, Statement 2 is false only for certain groups
 - (b) True, True
 - (c) Statement 1 is always false, Statement 2 is true only for certain groups
 - (d) True, False
 - (e) Statement 1 is true only for certain groups, Statement 2 is always true
4. Solve the equation $-6x = -36$. Check your answer.
 - (a) 42
 - (b) 30
 - (c) -42
 - (d) -36
 - (e) $\frac{1}{216}$
5. What is the order of group $\mathbb{Z}_{18}$?
 - (a) 30
 - (b) 15

- (c) 12
- (d) 6
- (e) 24

True / False (10 marks)

Answer True or False

6. True or False: Sixty percent of thirty is equal to one point eight.
7. True or False: The value $c = 1$ in $\mathbb{Z}_3$ ensures that $\mathbb{Z}_3[x]/(x^3 + x^2 + c)$ forms a field.
8. True or False: The polynomial is reducible over $\mathbb{Z}$, which contradicts its irreducibility over $\mathbb{Q}$.
9. True or False: The covariance function $C_N(2, 4)$ for a Poisson process with rate $= 4$ is equal to 14.
10. True or False: The solution to the equation $v - 26 = 68$ is $v = 102$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the roots of

$$6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0.$$

Enter the roots, separated by commas.

12. Discuss how the amount of money Leslie spent, specifically 20 dollars and 8 cents, can be represented as the fraction $20/8$. Analyze the implications of this representation in financial contexts.

End of Paper